

TalentSync – Technical Stack Document

Smart Recruitment Platform | Submission Document

1. Project Overview

TalentSync is a Smart Recruitment Platform implemented as a .NET 8 web application. The project contains an ASP.NET Core backend, a static HTML/CSS/JavaScript frontend, SQL Server database access through Entity Framework Core, JWT-based authentication and role-based authorization for Admin, Employer and JobSeeker users.

2. Core Technical Stack

Category	Technology used
.NET Version	.NET 8 (TargetFramework: net8.0)
Backend	ASP.NET Core Web API
Frontend	HTML5, CSS3 and JavaScript (static frontend; no frontend framework detected)
Database	Microsoft SQL Server / LocalDB (SmartRecruitmentDB)
ORM / Data Access	Entity Framework Core 8 with SQL Server provider
Programming Languages	C#, JavaScript, HTML, CSS
Authentication	JWT Bearer Authentication
Password Security	BCrypt.Net-Next
API Documentation	Swagger / OpenAPI using Swashbuckle.AspNetCore
Version Control	Git and GitHub

3. Backend Technologies and Libraries

The project file references the following NuGet packages:

- BCRYPT.Net-Next – 4.2.0
- Microsoft.AspNetCore.Authentication.JwtBearer – 8.0.29
- Microsoft.EntityFrameworkCore – 8.0.29
- Microsoft.AspNetCore.Identity.EntityFrameworkCore – 8.0.29
- Microsoft.EntityFrameworkCore.Design – 8.0.29
- Microsoft.EntityFrameworkCore.SqlServer – 8.0.29
- Microsoft.EntityFrameworkCore.Tools – 8.0.29
- Microsoft.IdentityModel.Tokens – 8.22.0
- Swashbuckle.AspNetCore – 6.6.2
- System.IdentityModel.Tokens.Jwt – 8.22.0

4. Database

The backend uses Entity Framework Core with Microsoft SQL Server. The configured connection string targets LocalDB and the database name is **SmartRecruitmentDB**. Entity Framework migrations are included in the project, including the initial database creation, job-matching changes, user activation change and remaining project tables.

Main database entities represented by ApplicationDbContext include:

- Users
- Employers
- Companies
- Jobs
- Applications
- ContactRequests
- JobSeekerProfiles
- JobSeekerSkills
- Educations
- Experiences
- CvDocuments
- Notifications

5. Authentication and Authorization

- JWT Bearer authentication is configured in Program.cs.
- Role-based authorization is used for Admin, Employer and JobSeeker controllers.
- User accounts are checked for IsActive status when JWT tokens are validated.
- Passwords are hashed using BCrypt.Net-Next.
- Registration is restricted to JobSeeker and Employer roles; Admin is seeded by the application.

6. Main Functional Modules

Module	Implemented functionality
Authentication	Register, Login, JWT token generation, authenticated test endpoints
Admin	Dashboard, user listing, user details and user activation/status management
Employer	Employer profile, company management, job creation/update/close, applicants and contact requests
JobSeeker	Profile, skills, education, experience, CV upload/delete and notifications
Job Matching	Job search, job details, match score, skill gap, apply and application history

7. Job Matching Technology

The project contains a configurable match-score service. The configured matching weights are **Skills 50%**, **Experience 25%**, **Education 15%** and **Location 10%**. The backend exposes job search, job details, application and skill-gap functionality for JobSeeker users.

8. API / Services Used

- Internal RESTful HTTP APIs exposed by ASP.NET Core controllers.
- Swagger/OpenAPI for API exploration and testing during development.

- JWT Bearer tokens for authenticated API requests.
- SQL Server database service through Entity Framework Core.

External third-party API integrations: No external API/service integration was identified in the uploaded project source. The project primarily uses its own backend REST APIs and local SQL Server database.

9. Frontend Structure

The frontend is served as static files by the ASP.NET Core application. It contains HTML pages and JavaScript/CSS modules for Authentication, Admin, Employer and JobSeeker areas.

Detected frontend pages include:

- Admin/dashboard.html
- Admin/profile.html
- Admin/settings.html
- Admin/user-details.html
- Admin/users.html
- Authentication/login.html
- Authentication/register.html
- Employer/applicants.html
- Employer/company.html
- Employer/dashboard.html
- Employer/job-form.html
- Employer/jobs.html
- JobSeeker/applications.html
- JobSeeker/cv.html
- JobSeeker/dashboard.html
- JobSeeker/engagement.html
- JobSeeker/job-details.html
- JobSeeker/jobs.html
- JobSeeker/profile.html
- JobSeeker/skills.html

10. Development Tools

- Visual Studio / .NET CLI for building and running the .NET project.
- SQL Server LocalDB and SQL Server tools for database management.
- Swagger UI for API testing.
- Git and GitHub for source control and repository submission.

11. Configuration Summary

Database: **SmartRecruitmentDB** using SQL Server LocalDB.

JWT issuer: **SmartRecruitmentPlatform**; audience: **SmartRecruitmentPlatformUsers**; token expiry: **60 minutes**.

Matching weights: Skills 50, Experience 25, Education 15, Location 10.

12. Source Basis

This document was prepared by inspecting the uploaded TalentSync/Smart Recruitment Platform source archive, including the project file, Program.cs, appsettings.json, database context, controllers, migrations and frontend files. Items not evidenced by the uploaded source have not been added as project technologies.

Security Note

The uploaded source currently contains a seeded Admin account and a JWT signing key in application configuration. Because the submission repository is required to be public, these credentials/secrets should ideally be moved to secure environment configuration and the temporary admin seed should be removed after the account is created. The credentials below are included only because the submission instructions require working login details.